Upper Chewaucan Watershed Restoration

Project (2025-2028)

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Introduction and Context

When doing research for the Contingent Valuation Project this term, I came across a project being processed for the Upper Chewaucan Watershed Restoration in Lake County, Oregon. The area covered by the project is in South-Central Oregon and covers the Chewaucan River and thirteen tributaries (rivers or streams flowing into a larger river). The impacts of this restoration project will not necessarily be felt directly from residents of Oregon or Lake County but will certainly impact local ranches as well as native species such as the Reband trout, mule deer, and waterfowl. The value derived from this project may exist in a wide variety of forms including both use and non-use values. The use values may be direct in the case of ranchers as well as fishers in the area. Non-use value may be derived from knowing that the habitat is being restored and that it will be there for the generations to come (Neider and Coordinator).

The restoration aims to address the various forms of habitat degradation that has historically occurred. Droughts, wildfires, grazing practices, and infrastructure development have all negatively impacted the ecosystem in the Upper Chewaucan Watershed (Neider and Coordinator).

In order to gauge the public support for this project in terms of Willingness to Pay (WTP) for the restoration, I designed a contingent valuation survey in order to collect data from people in Oregon. The goal was to get a variety of individuals to respond to the survey; however, there were limitations with certain groups not being as responsive as others. More details are listed in the Recruitment Strategy section below.

1) Contingent Valuation Survey

Recruitment Strategy/Limitations: Due to the budget constraint and time limitations of this survey, the respondents were selected based on convenience. I emailed my survey using a Google form to 60 individuals. In total, I received 31 responses. The issue with convenience sampling is that it introduces a bias to the study. Over and above the bias that comes with convenience, the people recruited tended to be more "outdoorsy" individuals and despite

them responding with their personal incomes, their actual family income might be much larger. The results cannot be generalized to the larger population.

Another issue comes from the size of the sample. With only 31 responses, the statistical power of the analysis is very limited. In an ideal setting the survey would achieve far more responses as well as randomization of respondents.

Background: The Lake Country Umbrella Watershed Council (hireafter "LCUWC") along with the Oregon Department of Fish and Wildlife (hireafter "ODFW") and the Oregon Watershed Enchancement Board, has proposed a project aimed at restoring the Upper Chewaucan Watershed in Lake County, Oregon. The focus of the project is on the following:

1.) Design fish passage improvement

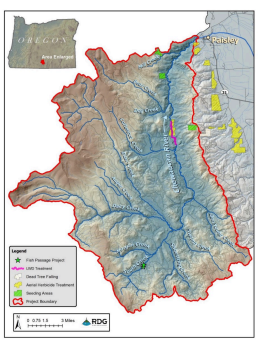
2.) Increase stream corridor habitat complexity

3.) Enhance rangeland

4.) Restore post-fire upland forest health

Goal: Improve the ecological function as well as the resilience of the Upper Chewaucan watershed

Ecological functionality in the Chewaucan watershed has faced degradation due to droughts, complications coming from grazing practices, infrastructure development, as well as wildfires burning much of the Chewaucan watershed.



The species impacted most by this degradation is the Redband trout which has been harmed through habitat fragmentation, cutting them off from 71% of their historical habitat. All the efforts will be focused on aquatic habitat restoration through the multi-pronged approach listed above ("Grant R25AP00230 Lake County Umbrella Watershed Council").

Solution proposal: The projects study and design phase has been awarded federal support but additional funding will be necessary. The goal is to raise funding through a state tax

focused on restoration efforts in the Chewaucan Watershed. The funds will be used to implement the restoration project over a 3 year period from 2025-2028. Would you be willing to pay an additional ____ per year in state taxes to fund the restoration of the Upper Chewaucan Watershed?

Randomized Bids

a) $10
b) $15
c) $20
d) $30
e) $45

(Answer in the case of zero bid) : Did you reply "No" because:

a) The change the program may create is not significant
b) You cannot afford to pay more taxes
c) You do not support the program
d) The project does not impact you
e) Other: _____

Demographic Questions:

1. City of Residence: _____

2. Do you live in Oregon full-time?

a) Yes

b) No

3. What is you annual household income?

a) Less than $25,000

b) $25,000 - $50,000

c) $50,000 - $100,000

d) $100,000 - $150,000

e) More than $150,000

4. Have you traveled around South-Central Oregon?

a) Yes

b) No

c) Unsure

5. Do you visit Oregon rivers, lakes, or parks?

a) Yes

b) No

Results

In order to calculate the average Willingness to Pay (WTP hitherafter) for the Upper Chewaucan Watershed Restoration project, we utilized a logit regression model as specified in Hanemann (1984). The model estimates a bid function which explains how WTP varies with alterations in the price of the good (the tax increase in this case) as well as other characteristics of the respondents (Bateman et al.).

When calculating the average WTP we used the derived formula:

$$WTP = \frac{ConstantCoef + (Controls * ControlCoefs)}{|PriceCoef|}$$

$$Coef = \ Coefficients\ of\ variables$$

$$Controls = \ Mean\ values\ of\ control\ variables$$

Results from Logit Regression

Logit Regression Note: Lower income category is the reference group for income bin

Table 1: Effects from Logit Model (glm)

	(1)
Constant	-0.083
	(0.963)
Bid ($)	-0.077
	(0.052)
Income: Middle ($50k–$150k)	1.816
	(0.197)

	(1)
Income: High (>$150k)	0.495
	(0.785)
Lives in Oregon full-time	0.956
	(0.369)
Traveled in South-Central Oregon	0.933
	(0.413)
Visits Rivers Lakes or Parks	1.425
	(0.372)
Num.Obs.	31
AIC	44.2
BIC	54.3
Log.Lik.	-15.110
F	0.976
RMSE	0.40

Table 2: Marginal Effects from Logit Model (logitmfx)

	(1)
Bid ($)	-0.015
	(0.008)
Income: Middle ($50k–$150k)	0.276
	(0.156)
Income: High (>$150k)	0.088
	(0.284)
Lives in Oregon full-time	0.201
	(0.227)
Traveled in South-Central Oregon	0.198
	(0.254)
Visits Rivers Lakes or Parks	0.329
	(0.380)

Interpretation of Marginal Effects

The marginal effects calculated above describe the change in probability of accepting the presented bid amount for a one unit increase in the independent variable (in context of bid amount) and for the binary variables it describes the change in probability of accepting the bid when moving from yes (1) to no (0).

In this case, the price coefficient describes that for every additional dollar increase, the probability of accepting the proposed bid decreases by 1.5 percentage points ceteris paribus. For income (middle), income (high), living in Oregon full-time, traveling in South-Central

Oregon, and visiting rivers/lakes/parks, the probability of accepting the bid increases by 27.6, 8.8, 20.1, 19.8, and 32.9 respectively ceteris paribus. It is necessary to note that these results are not statistically significant.

Mean Willingness to Pay (WTP) Estimate

Based on the results from the logit regression, we estimate that the average Willingness to Pay (WTP) for the Upper Chewaucan Watershed Restoration project is approximately **$38.50** per year in additional state taxes. This estimate takes into account various demographic factors such as income level, residency, and recreational habits.

Stress Testing with Protest Zero Adjustments

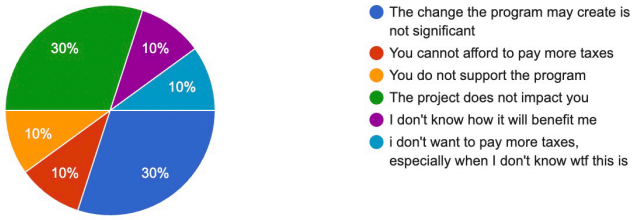
Protest zeros are specified as responses in which the person has a non-zero WTP but answer "No" to the specified bid amount. Carson (2000) describes how "a variety of explanations for protest zero ranging from rejection of the legitimacy of the scenario presented to strategic behavior have been put forth. When asking discrete choice questions, the corresponding phenomenon is sometimes labeled nay-saying. This occurs when the respondent provides a no response to an amount asked even though WTP is greater than the amount asked about. The presence of "untruthful" responses, for whatever reason, leads to arguments that contingent valuation responses should be "calibrated" to potentially correct for either an upward or downward bias".

To account for the potential bias that may arise from protest zeros in the analysis, I included the following screening question:

Did you reply "No" because...

a) The change the program may create is not significant
b) You cannot afford to pay more taxes
c) You do not support the program
d) The project does not impact you
e) Other: _____

Results:



Based on the definition outlined in Carson(2000), the responses that are protest zeros are when respondents answered:

1. You cannot afford to pay more taxes
2. You do not support the program
3. Other: "i don't want to pay more taxes, especially when I don't know wtf this is"

Table 3: Effects from Logit Model (glm) - Protest Zeros Removed

	(1)
Constant	0.556
	(0.770)
Bid ($)	-0.074
	(0.103)
Income: Middle ($50k–$150k)	1.802
	(0.255)
Income: High (>$150k)	-0.618
	(0.780)
Lives in Oregon full-time	0.720
	(0.568)
Traveled in South-Central Oregon	-0.371
	(0.821)
Visits Rivers Lakes or Parks	2.476
	(0.157)
Num.Obs.	28
AIC	37.2
BIC	46.6

	(1)
Log.Lik.	-11.612
F	0.940
RMSE	0.36

Table 4: Marginal Effects from Logit Model (logitmfx) - Protest Zeros Removed

	(1)
Bid ($)	-0.011
	(0.007)
Income: Middle ($50k–$150k)	0.205
	(0.134)
Income: High (>$150k)	-0.108
	(0.441)
Lives in Oregon full-time	0.116
	(0.216)
Traveled in South-Central Oregon	-0.052
	(0.216)
Visits Rivers Lakes or Parks	0.524
	(0.359)

In this case, the price coefficient describes that for every additional dollar increase, the probability of accepting the proposed bid decreases by 1.1 percentage points ceteris paribus. For income (middle), income (high), living in Oregon full-time, traveling in South-Central Oregon, and visiting rivers/lakes/parks, the probability of accepting the bid increases by 20.5, decreases by 10.8, increases by 11.6, decreases by 5.2, and increases by 52.4 percentage points respectively ceteris paribus. It is necessary to note that these results are not statistically significant.

After removing three protest zeros from the analysis the mean WTP is estimated to be **$48.29** per year in additional state taxes.

References

Bateman, Ian, et al. Economic Valuation with Stated Preference Techniques: A Manual.

Carson, Richard T. "Contingent Valuation: A User's Guide." Environmental Science & Technology, vol. 34, no. 8, Apr. 2000, pp. 1413–1418, www.econ.ucsd.edu/~rcarson/papers/CVusersguide.pdf, https://doi.org/10.1021/es990728j.

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AI use acknowledgement: I used AI tools to help with various LaTeX formatting issues. It helped design tables to better summarize my data. Furthermore, I used AI tools for data cleansing mainly using co-pilot. Much of what AI provided was edited and reworked to fit my needs. When calculating the mean WTP for the project I attempted the code and troubleshot with AI when needed. Furthermore, AI was used to help grammar check and rephrase for clarity in various sections.